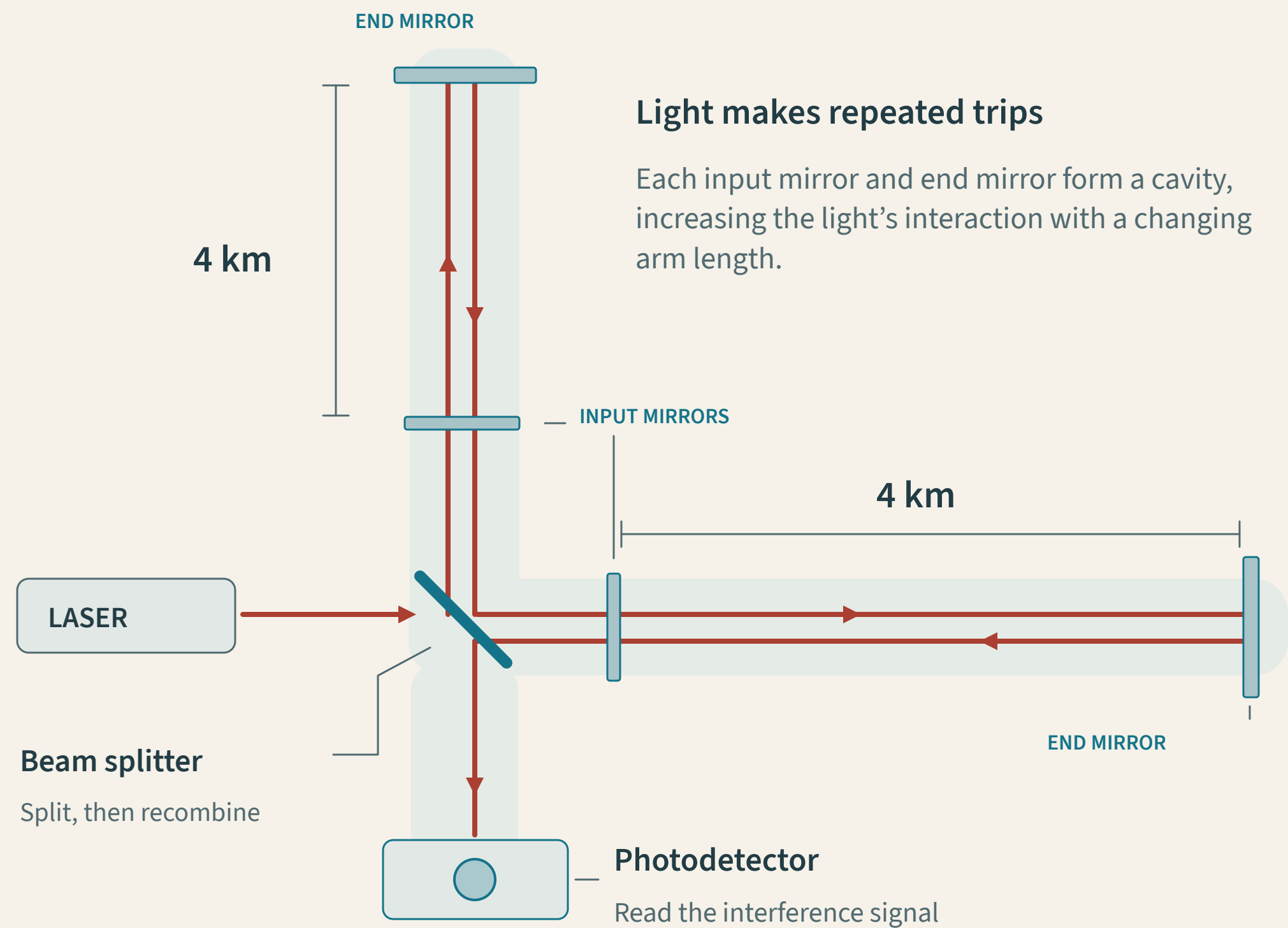


LIGO: turning a change in space into light

Two long optical cavities amplify a tiny differential effect; interference makes the change observable.

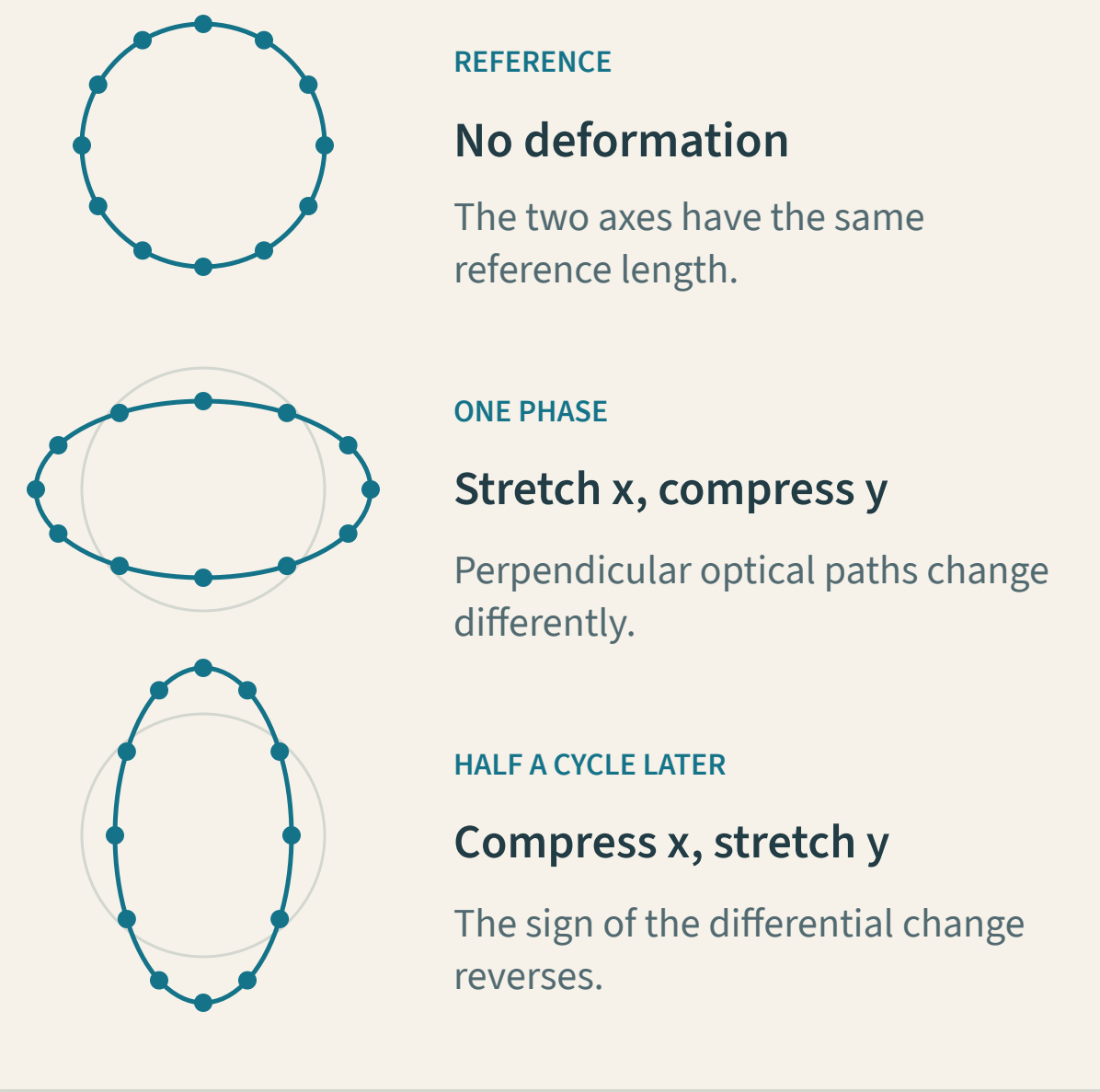
01 Follow the light

A Michelson layout with two perpendicular Fabry–Pérot arm cavities.



02 Space changes the paths

A conceptual ring of freely falling test particles, viewed along the wave.

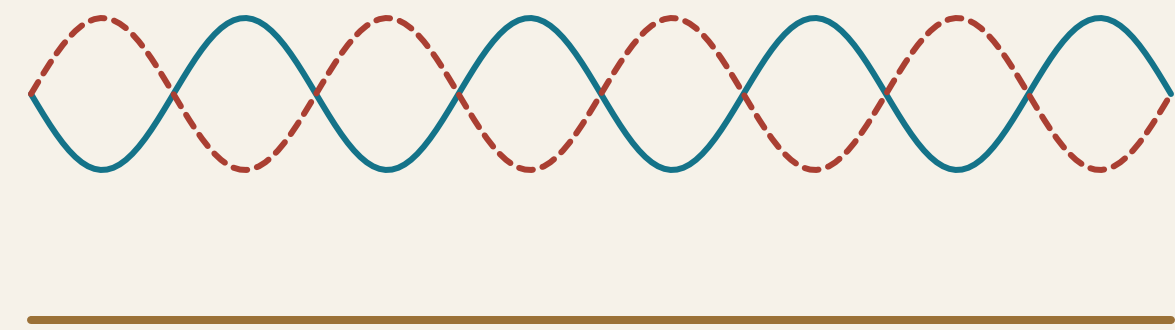


Plus polarization shown. Deformation is greatly exaggerated; the faint reference is the original circle.

03 Compare optical phase, not a ruler reading

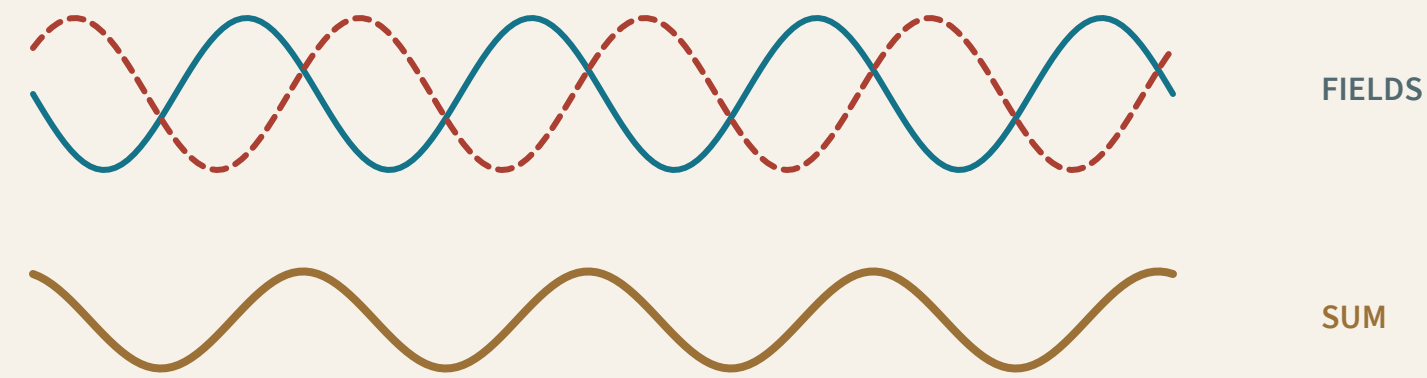
Returning electric fields combine. A relative phase change alters the light reaching the output.

IDEAL DESTRUCTIVE INTERFERENCE



Opposite fields cancel at the dark output.

WITH A SMALL PHASE DIFFERENCE



Incomplete cancellation changes output power.